Merge Two Circular Singly Linked List

```
#include <stdio.h>
#include <stdlib.h>

struct node
{
    int data;
    struct node *next;
};

/* Create Circular Linked List */
struct node* createList(int n)
{
    struct node *head=NULL,*temp=NULL,*newnode;
    int i,value;

    for(i=1;i<=n;i++)
    {
        newnode=(struct node*)malloc(sizeof(struct
node));

        printf("Enter value: ");
        scanf("%d",&value);

        newnode->data=value;

        if(head==NULL)
        {
            head=newnode;
            newnode->next=head;
            temp=newnode;
        }
        else
        {
            temp->next=newnode;
            newnode->next=head;
            temp=newnode;
        }
    }
    return head;
}
```

```
/* Display CSLL */
void display(struct node *head)
{
    struct node *temp;

    if(head==NULL)
    {
        printf("List empty\n");
        return;
    }

    temp=head;

    do
    {
        printf("%d -> ",temp->data);
        temp=temp->next;
    }while(temp!=head);

    printf("(back to head)\n");
}

/* Merge Two CSLL */
struct node* mergeList(struct node *head1, struct node
*head2)
{
    struct node *temp1,*temp2;

    if(head1==NULL) return head2;
    if(head2==NULL) return head1;

    temp1=head1;
    temp2=head2;

    /* find last node of first list */
    while(temp1->next!=head1)
        temp1=temp1->next;

    /* find last node of second list */
    while(temp2->next!=head2)
        temp2=temp2->next;

    /* connect both lists */
    temp1->next=head2;
    temp2->next=head1;

    return head1;
```

```
}

/* Main */
int main()
{
    struct node *head1=NULL,*head2=NULL,*merged=NULL;
    int n1,n2;

    printf("Enter size of first circular list: ");
    scanf("%d",&n1);

    head1=createList(n1);

    printf("Enter size of second circular list: ");
    scanf("%d",&n2);

    head2=createList(n2);

    merged=mergeList(head1,head2);

    printf("\nMerged Circular Linked List:\n");
    display(merged);

    return 0;
}
```